

Quaternion-Based Adaptive Backstepping RFWNN Control of Quadrotors Subject to Model Uncertainties and Disturbances

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Abstract This paper presents a quaternion-based adaptive backstepping control method using recurrent fuzzy wavelet neural network (RFWNN) for regulation and trajectory tracking of quadrotors subject to model uncertainties and disturbances. For the controller synthesis, a more complete model of an uncertain quadrotor is first obtained by incorporating with mass variations and wind disturbances, which are online learned by using the RFWNN. Afterward, a quaternion-based adaptive backstepping RFWNN controller is synthesized by integrating backstepping, quaternion control, and the RFWNN online learner. The closed-loop stability of the overall quadrotor control system is shown semi-globally uniformly ultimately bounded via Lyapunov stability theory. The effectiveness and performance of the proposed control method are well exemplified by conducting four simulations on hovering and three-dimensional sinusoidal trajectory tracking control of a quadrotor. Through the simulation results, the proposed control method is shown superior by comparing to two existing methods.

Keywords Backstepping · Quaternion · Quadrotor · Recurrent fuzzy wavelet neural network (RFWNN) · Regulation · Trajectory tracking

1 Introduction

Recently, quadrotors have become more and more popular due to not only their versatility and many commercial and civil applications, but also their good properties like low cost, simple structure, and easy control [1]. Quadrotors have gained many applications including bridge diagnosis, close-up inspection, detailed area mapping, agriculture, environmental monitoring and protection, and mineral exploration and exploitation. In order to meet the requirements of these missions and a variety of payloads [2, 3], conventional linear PID control may not be able to give required performance in the presence of large nonlinear dynamics and model uncertainties over wide operational ranges. The authors in [4–7] illustrated the examples of modeling uncertainties including mass changes, random winds, and variations of the center of gravity. Although the authors in [4–6] used PID or LQR controller to achieve disturbance rejection, the authors in [7] dealt with the disturbances by a switch control technique with adaptively learning the bound of the disturbances. Partially backstepping-like feedback linearization in [8] dealt with nonlinearities of the quadrotor, but the authors in [8] did not consider the uncertainties and disturbances. On the other hand, the integral backstepping method was used in [9] to show the ability of disturbance rejection.

In order to cope with significant model uncertainties and disturbances, nonlinear backstepping controller augmented with adaptive or disturbance rejection algorithm has shown the ability to deal with the uncertainties in [10, 11], but the Newton–Euler dynamic equations used in those papers have a vulnerability that is the singularity at $\pm \pi/2$ of pitch angle. Therefore, quaternion-based controllers were designed to overcome the singularity. The studies in [3, 12] proposed quaternion-based backstepping control methods

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for quadrotors without uncertainties. In [13–15], the quaternion-based backstepping controllers with adaptive algorithm or uncertainty observer were developed to deal with uncertainties. In particular, the adaptive model presented in [13] was linear in parameters, the study in [14] proposed a feedback linearization method with a reduced-order linear extended state observer (RLESO), and meanwhile, the method presented in [15] used a hyperbolic tangent function to model the uncertainties. Obviously, the authors in [12–15] showed the feasibility of the quaternion-based backstepping controller for quadrotors, but only the attitude control was achieved in [14]. However, as authors' best understanding, no investigations have been paid to apply the quaternion-based adaptive backstepping design method augmented with the RFWNN to approximate uncertainties for quadrotor control, in order to not only avoid the chattering effects caused by the switch control approaches in [2, 7], but also eliminate the singularity problems in [2, 4–11].

Adaptive fuzzy backstepping control for uncertain nonlinear system has been well studied by the authors in [16–18]; those researchers showed the merits on model approximations with fuzzy systems. Moreover, recurrent fuzzy wavelet neural networks (RFWNNs) have been used for nonlinear controller design, nonlinear identification, and function approximation [19]. Instead of neglecting the uncertain terms in [4–6, 8, 12], or replacing the linear observer in [14] as well as and the hyperbolic tangent functions in [15], RFWNNs will be used in this paper to approximate the uncertainties such that the full rigid body model can be considered for the controller design.

The objective of this paper is to design an adaptive quaternion-based backstepping flight control law augmented by RFWNN adaptive algorithm for quadrotors. By comparing the state of art in [8, 13], the proposed method is more effective and superior in deriving such an intelligent adaptive control law to achieve stabilization and trajectory tracking in the presence of significantly exogenous wind gusts and model uncertainties by mass variations. Overall, the novelty and contributions of this paper are stated as follows.

First, the flight control problems of nonlinearities, singularities, uncertainties, and disturbances for quadrotors are simultaneously addressed by the quaternion-based backstepping control law augmented by RFWNN adaptive algorithm. The proposed control law is shown capable of carrying out superior stable regulation and trajectory tracking performance than the methods in [8, 13] do.

Second, to simplify calculations of the backstepping virtual commands, the concept of the first-order Lyapunov-based command filter is introduced to derive the proposed control law rigorously. The closed-loop stability of the proposed intelligent adaptive control system is shown

semi-global ultimate uniformly bounded via the Lyapunov stability theory.

Third, by including the estimation errors of the RFWNN in the stability analysis, the proposed online supervised learning rules of the RFWNN are achieved without offline pre-learning. Furthermore, the learning process can be optimized by appropriately adjusting different ratios of the tracking errors and approximation errors in the error terms.

The rest of this paper is organized as follows. Section 2 states the modeling and problem formulation. In Sect. 3, the proposed quaternion-based backstepping RFWNN control law is designed in detail. Section 4 presents simulation results to show the effectiveness and superiority of the proposed method. Section 5 concludes this paper.

2 Modeling and Problem Formulation

In this section, the properties of quaternion mathematics will be first summarized, and then, a quaternion-based dynamic model of a quadrotor with uncertainties will be described.

2.1 Properties of the Quaternion Operation

The quaternion vector is expressed as:

$$\mathbf{q} = [\eta \ \boldsymbol{\varepsilon}^T]^T = [\cos(\theta/2) \ \sin(\theta/2)\hat{\mathbf{k}}^T]^T \quad (1)$$

where the unit vector $\hat{\mathbf{k}}$ describes the rotation axis and θ is the angle of rotation. Additionally, the inverse of the quaternion is $\mathbf{q}^{-1} = [\eta - \boldsymbol{\varepsilon}^T]^T$ and any rotational vector can be represented as a quaternion by using $\mathbf{q}(\boldsymbol{\omega}) = [0 \ \boldsymbol{\omega}^T]^T$. After the definition of the quaternion, the relationship of the direction cosine matrix and quaternion is expressed by

$$\mathbf{R}_b^I(\mathbf{q}) = \mathbf{I}_3 + 2\eta\mathbf{S}(\boldsymbol{\varepsilon}) + 2\mathbf{S}(\boldsymbol{\varepsilon})^2 \quad (2)$$

whose inverse matrix is denoted by $\mathbf{R}_I^b(\mathbf{q}) = \mathbf{R}_b^I(\mathbf{q})^T = \mathbf{I}_3 - 2\eta\mathbf{S}(\boldsymbol{\varepsilon}) + 2\mathbf{S}(\boldsymbol{\varepsilon})^2$. where the subscripts b and I, respectively, mean the body and inertial reference frame, and

$$\mathbf{S}(\boldsymbol{\varepsilon}) = \begin{bmatrix} 0, & -\varepsilon_3, & \varepsilon_2 \\ \varepsilon_3, & 0, & -\varepsilon_1 \\ -\varepsilon_2, & \varepsilon_1, & 0 \end{bmatrix}.$$

Therefore, multiplying $\mathbf{R}_b^I(\mathbf{q})$ to a vector will map the representation of this vector from body frame to inertial frame. Now, the quaternion production is defined by

$$\mathbf{q}_1 \circ \mathbf{q}_2 = \begin{bmatrix} \eta_1\eta_2 - \boldsymbol{\varepsilon}_1 \cdot \boldsymbol{\varepsilon}_2 \\ \eta_1\boldsymbol{\varepsilon}_2 + \eta_2\boldsymbol{\varepsilon}_1 - \boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2 \end{bmatrix} \quad (3)$$

From (2) and (3), the following summarizes the equivalence of rotating a vector $\mathbf{v}$

$$\begin{cases} q(v^i) = q(R_b^i(q)v^b) = q^{-1} \circ q(v^b) \circ q \\ q(v^b) = q(R_i^b(q)v^i) = q \circ q(v^i) \circ q^{-1} \end{cases} \quad (4)$$

$$\begin{cases} R_b^i(q_2)R_b^i(q_1) = R_b^i(q_1 \circ q_2) \\ R_i^b(q_2)R_i^b(q_1) = R_i^b(q_2 \circ q_1) \end{cases} \quad (5)$$

2.2 Quadrotor Modeling with Uncertainties

This section describes the model of a quadrotor. Although effects of different kinds of uncertainties were discussed in [4–7], the model in this paper will simultaneously consider several kinds of uncertainties, in order to derive our proposed control law later on. As depicted in Fig. 1, the fourth-order dynamic state equations of a quadrotor are described as follows;

$$\dot{P} = v \quad (6)$$

$$\dot{v} = g\hat{e}_3 - \frac{T}{\hat{m} + \Delta m} R_b^l(q)\hat{e}_3 + \frac{1}{\hat{m} + \Delta m} F_a \quad (7)$$

$$\dot{q} = \frac{1}{2} q(\omega^b) \circ q \quad (8)$$

$$(\hat{J} + \Delta J)\dot{\omega}^b = -S(\omega^b)(\hat{J} + \Delta J)\omega^b + \tau + \tau_t + \tau_g + \tau_\omega + \tau_a \quad (9)$$

where $\hat{m}$ is the estimation of the quadrotor mass and Δm is the mass variation. In addition, $\tau = \sum_{i=1}^4 r_i \times F_i - \sum_{i=1}^4 Q_i$ denotes the control torque; the differences of control torques to real torque generated by motors are $\tau_t = \sum_{i=1}^4 \Delta r_i \times F_i + \sum_{i=1}^4 \Delta Q_i$; the moment due to the variation of the position of center of gravity is $\tau_g = \Delta x_{cg} \times mg R_b^b(q)\hat{e}_3$; $\tau_\omega = \sum_{i=1}^4 (-1)^{i+1} I_m(\omega \times \hat{e}_3)\omega_i$ is the thruster gyroscopic moment; the thrusters' arm vectors are $r_1 = [l \ 0 \ 0]^T$, $r_2 = [0 \ l \ 0]^T$, $r_3 = [-l \ 0 \ 0]^T$, and $r_4 = [0 \ -l \ 0]^T$; the thrust force and thruster moment are $F_i = -k\omega_i^2 \hat{e}_3$ and $Q_i = (-1)^i b\omega_i^2 \hat{e}_3$, respectively. F_a and τ_a are forces and torques generated by wind. Moreover, the relationship between the motor angular speeds and the control inputs is expressed in the following equation.

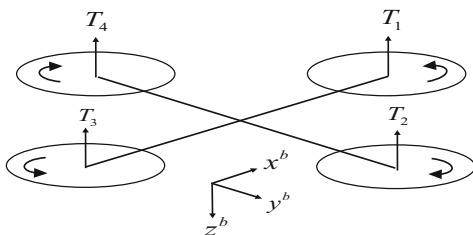


Fig. 1 Axis definitions of the quadrotor

$$\begin{bmatrix} T \\ \tau \end{bmatrix} = \begin{bmatrix} k & k & k & k \\ 0 & -lk & 0 & lk \\ lk & 0 & -lk & 0 \\ b & -b & b & -b \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \quad (10)$$

3 Proposed Method

3.1 General form of the Quadrotor Model

In order to use RFWNN to approximate the uncertainties, it is necessary to transform the motion of equations in (6)–(9) to a more general form. By rewriting (7) and (9), we have

$$\begin{aligned} \dot{v} &= g\hat{e}_3 - \frac{T}{\hat{m}} R_b^l(q)\hat{e}_3 + \left(\frac{T}{\hat{m}} - \frac{T}{\hat{m} + \Delta m}\right) R_b^l(q)\hat{e}_3 + \frac{1}{\hat{m} + \Delta m} F_a \\ &= g\hat{e}_3 - \frac{T}{\hat{m}} R_b^l(q)\hat{e}_3 + \Delta_a \end{aligned} \quad (11)$$

$$\begin{aligned} \dot{\omega}^b &= \hat{J}^{-1}[-S(\omega^b)\hat{J}\omega^b + \tau] \\ &\quad + \hat{J}^{-1}[-S(\omega^b)\Delta J\omega^b - \Delta J\dot{\omega}^b + \tau_t + \tau_g + \tau_\omega + \tau_a] \\ &= \hat{J}^{-1}[-S(\omega^b)\hat{J}\omega^b + \tau] + \Delta_x \end{aligned} \quad (12)$$

which gives the uncertain model of the quadrotor as

$$\dot{P} = v \quad (13)$$

$$\dot{v} = g\hat{e}_3 - \frac{T}{\hat{m}} R_b^l(q)\hat{e}_3 + \Delta_a \quad (14)$$

$$\dot{q} = \frac{1}{2} q(\omega^b) \circ q \quad (15)$$

$$\dot{\omega}^b = \hat{J}^{-1}[-S(\omega^b)\hat{J}\omega^b + \tau] + \Delta_x \quad (16)$$

where Δ_a and Δ_x are the total effects caused by the uncertainties.

3.2 RFWNN Function Approximation

Figure 2 depicts a schematic diagram of the proposed RFWNN structure composed of four layers: input, membership, rule, and output, each of which has the subsequent function.

(1) Input layer:

All nodes of the input layer directly transfer input signals to the next layer. There are totally m input nodes, and the input signal can be represented as a vector form, $x = [x_1, \dots, x_m]^T \in R^m$.

(2) Membership layer:

The membership layer has totally M neuron group, each of which is with m nodes. Each node performs its corresponding membership function to the only-one input. The

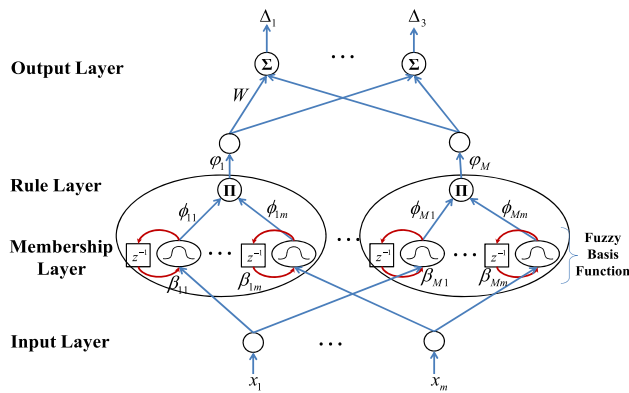


Fig. 2 Structure of the RFWNN

input of the j th node in i th neuron group is the sum of the output from the j th node of the previous layer and the output from itself through a time delay. The membership layer adopts the following wavelet function as its membership function.

$$\phi_{ij} = \left[1 - \alpha_{ij}^2 (\bar{x}_{ij} - c_{ij})^2 \right] e^{-\alpha_{ij}^2 (\bar{x}_{ij} - c_{ij})^2} \quad (17)$$

where $i = 1, \dots, M$, $j = 1, \dots, m$; $\bar{x}_{ij}$, c_{ij} , and $1/\alpha_{ij}^2$ are, respectively, the input, mean, and variance of the wavelet function of j th node in the i th neuron group. Moreover, $\bar{x}_i(k) = x(k) + \text{diag}(\beta_i)\phi_i(k-1)$ and $\beta_i \in R^{1 \times m}$ denote the recurrent constant parameter of the i th neuron; $\phi_i = [\phi_{i1}, \dots, \phi_{im}]^T$, $\bar{x}_i = [\bar{x}_{i1}, \dots, \bar{x}_{im}]^T$. The dimensions of the parameters are defined by $\alpha, \beta, c \in R^{M \times m}$.

(3) Rule layer:

Each node of the rule layer multiplies all of its input signals and obtains the subsequent function. $\varphi_i = \prod_{j=1}^m \phi_{ij}$.

(4) Output layer:

The node outputs in the output layer are the weighted sum of their input signals such that the outputs of the RFWNN are described by

$$\Delta = W^T \varphi \quad (18)$$

where $W \in R^{M \times N}$, $\varphi \in R^M$; N is the number of nonlinear functions; $\varphi = [\varphi_1, \dots, \varphi_M]^T$. Let the ideal approximation result be

$$\Delta^* = W^{*T} \varphi^* + \varepsilon^* \quad (19)$$

where W^* is the optimal weight vector; φ^* is the optimal rule-layer output vector; ε^* is a small and bounded error vector. Note that the norms of the optimal weights $\|W^*\|$, $\|\alpha^*\|$, $\|\beta^*\|$, and $\|c^*\|$ are bounded. Hence, the actual fuzzy

wavelet network estimation, $\hat{\Delta}$, and parameters $\hat{W}$, $\hat{\alpha}$, $\hat{\beta}$, and $\hat{c}$ are estimates of Δ^* , W^* , α^* , β^* , and c^* . By defining $\tilde{\Delta} = \Delta^* - \hat{\Delta}$, $\tilde{W} = W^* - \hat{W}$, $\tilde{\alpha} = \alpha^* - \hat{\alpha}$, $\tilde{\beta} = \beta^* - \hat{\beta}$, and $\tilde{c} = c^* - \hat{c}$, (19) can be rewritten as

$$\Delta^* = \hat{\Delta} + \tilde{W}^T \tilde{\varphi} + \tilde{W}^T \hat{\varphi} + \tilde{W}^T \tilde{\varphi} + \varepsilon^* \quad (20)$$

Expand $\tilde{\varphi} = \varphi^* - \hat{\varphi}$ by Taylor series as

$$\tilde{\varphi} = \frac{\partial \varphi}{\partial \alpha} \tilde{\alpha} + \frac{\partial \varphi}{\partial \beta} \tilde{\beta} + \frac{\partial \varphi}{\partial c} \tilde{c} + \tilde{\varepsilon} \quad (21)$$

Thus,

$$\begin{aligned} \tilde{\Delta} &= \Delta^* - \hat{\Delta} \\ &= \tilde{W}^T \hat{\varphi} + \tilde{W}^T \frac{\partial \varphi}{\partial \alpha} \tilde{\alpha} + \tilde{W}^T \frac{\partial \varphi}{\partial \beta} \tilde{\beta} + \tilde{W}^T \frac{\partial \varphi}{\partial c} \tilde{c} + \varepsilon \end{aligned} \quad (22)$$

where $\varepsilon = \tilde{W}^T \tilde{\varphi} + \varepsilon^* + \tilde{\varepsilon}$ and ε is assumed bounded, i.e., $\|\varepsilon\|_\infty < \varepsilon_{\max} < \infty$.

3.3 Controller Design

The design of the proposed quaternion-based adaptive backstepping RFWNN controller is proposed in this subsection. The control law is designed in the following four steps. The closed-loop stability is analyzed in Sect. 3.4 and is finally concluded by Theorem 1.

Step 1 The position error and its time derivative are defined by

$$\delta_1 = P - P_d, \quad \dot{\delta}_1 = v - \dot{P}_d \quad (23)$$

Let the proposed Lyapunov function be $V_1 = 0.5\delta_1^T \delta_1$. By differentiating the Lyapunov function, one obtains $\dot{V}_1 = \delta_1^T (v - \dot{P}_d)$. Define the virtual control law

$$v = \Phi_1 = \dot{P}_d - k_1 \delta_1 \quad (24)$$

Then, $\dot{V}_1 = -k_1 \delta_1^T \delta_1$ which is negative definite.

Step 2 Define the velocity backstepping error by

$$\delta_2 = v - \Phi_1 \quad (25)$$

Substituting (25) into (23) obtains

$$\dot{\delta}_1 = -k_1 \delta_1 + \delta_2 \quad (26)$$

The derivative of velocity backstepping error is

$$\dot{\delta}_2 = g \hat{e}_3 - \frac{T}{\hat{m}} R_b^l(q) \hat{e}_3 + \Delta_a - \ddot{P}_d - k_1^2 \delta_1 + k_1 \delta_2 \quad (27)$$

Define the Lyapunov function $V_2 = V_1 + 0.5\delta_2^T \delta_2$ and use (26) and (27) to obtain

$$\dot{V}_2 = -k_1 \delta_1^T \delta_1 + \delta_2^T \left(\delta_1 + g \hat{e}_3 - \frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 + \Delta_a - \ddot{\mathbf{P}}_d - k_1^2 \delta_1 + k_1 \delta_2 \right)$$

We then propose the virtual control law as

$$\begin{aligned} -\frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 &= -\frac{T_d}{\hat{m}} \mathbf{R}_b^l(q_d) \hat{e}_3 \\ &= \Phi_2 = -g \hat{e}_3 - \hat{\Delta}_a + \ddot{\mathbf{P}}_d + (k_1^2 - 1) \delta_1 \\ &\quad - (k_1 + k_2) \delta_2 \end{aligned} \quad (28)$$

such that $\dot{V}_2 = -k_1 \delta_1^T \delta_1 - k_2 \delta_2^T \delta_2 + \delta_2^T \tilde{\Delta}_a$.

Step 3 The backstepping error of the desired acceleration vector with the actual acceleration vector is defined by

$$\delta_3 = -\frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 - \Phi_2 \quad (29)$$

Next, we introduce a first-order command filter in the sense of Lyapunov stability into the backstepping control design. We will show that (29) can be further decomposed into three error vectors. By rewriting (29), we have

$$\begin{aligned} \delta_3 &= -\frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 - \Phi_2 = -\frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 + \frac{T_d}{\hat{m}} \mathbf{R}_b^l(q_d) \hat{e}_3 \\ &= -\frac{T}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 + \frac{T_{df}}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 - \frac{T_{df}}{\hat{m}} \mathbf{R}_b^l(q) \hat{e}_3 \\ &\quad + \frac{T_{df}}{\hat{m}} \mathbf{R}_b^l(q_{df}) \hat{e}_3 - \frac{T_{df}}{\hat{m}} \mathbf{R}_b^l(q_{df}) \hat{e}_3 + \frac{T_d}{\hat{m}} \mathbf{R}_b^l(q_{df}) \hat{e}_3 \\ &\quad - \frac{T_d}{\hat{m}} \mathbf{R}_b^l(q_{df}) \hat{e}_3 + \frac{T_d}{\hat{m}} \mathbf{R}_b^l(q_d) \hat{e}_3 \\ &= \left(-\frac{T}{\hat{m}} + \frac{T_{df}}{\hat{m}} \right) \mathbf{R}_b^l(q) \hat{e}_3 - \frac{T_{df}}{\hat{m}} [\mathbf{R}_b^l(q) - \mathbf{R}_b^l(q_{df})] \hat{e}_3 \\ &\quad - \frac{T_d}{\hat{m}} [\mathbf{R}_b^l(q_{df}) - \mathbf{R}_b^l(q_d)] \hat{e}_3 + \left(-\frac{T_{df}}{\hat{m}} + \frac{T_d}{\hat{m}} \right) \mathbf{R}_b^l(q_{df}) \hat{e}_3 \end{aligned}$$

and letting $T = T_{df}$ with the following three definitions

$$\delta_{3a} = -\frac{T_{df}}{\hat{m}} [\mathbf{R}_b^l(q) - \mathbf{R}_b^l(q_{df})] \hat{e}_3 \quad (30)$$

$$\delta_{3b} = -\frac{T_d}{\hat{m}} [\mathbf{R}_b^l(q_{df}) - \mathbf{R}_b^l(q_d)] \hat{e}_3 \quad (31)$$

$$\delta_{3c} = -\frac{T_{df}}{\hat{m}} + \frac{T_d}{\hat{m}} \quad (32)$$

Hence, (29) turns out

$$\delta_3 = \delta_{3a} + \delta_{3b} + \delta_{3c} \mathbf{R}_b^l(q_{df}) \hat{e}_3 \quad (33)$$

where T_d is the desired thrust command; T_{df} is the filtered thrust command; q_d and q_{df} represent desired orientation command and the orientation command after filtering, respectively. One substitutes (28) and (33) into (27) and then finds

$$\dot{\delta}_2 = -\delta_1 - k_2 \delta_2 + \delta_{3a} + \delta_{3b} + \delta_{3c} \mathbf{R}_b^l(q_{df}) \hat{e}_3 + \tilde{\Delta}_a \quad (34)$$

Before continuing the backstepping process, we need the following definition and Lemma to reformulate (30) and (31).

Definition 1 Define

$$\tilde{q} = [\tilde{\eta} \tilde{\varepsilon}^T]^T = q_{df}^{-1} \circ q \quad (35)$$

$$\tilde{q}_{df} = [\tilde{\eta}_{df} \tilde{\varepsilon}_{df}^T]^T = q_d^{-1} \circ q_{df} \quad (36)$$

Consequently, if $q = q_{df}$, then $\tilde{q} = [1 \ 0^T]^T$; if $q_{df} = q_d$, then $\tilde{q}_{df} = [1 \ 0^T]^T$.

Lemma 1 Following the definitions of (35) and (36) gives

$$\dot{\tilde{q}} = \begin{bmatrix} \dot{\tilde{\eta}} \\ \dot{\tilde{\varepsilon}} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\tilde{\varepsilon}^T \\ \tilde{\eta} \mathbf{I} + S(\tilde{\varepsilon}) \end{bmatrix} \tilde{\omega} \quad (37)$$

$$\dot{\tilde{q}}_{df} = \begin{bmatrix} \dot{\tilde{\eta}}_{df} \\ \dot{\tilde{\varepsilon}}_{df} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\tilde{\varepsilon}_{df}^T \\ \tilde{\eta}_{df} \mathbf{I} + S(\tilde{\varepsilon}_{df}) \end{bmatrix} \tilde{\omega}_{df} \quad (38)$$

where $\tilde{\omega} = \mathbf{R}_b^l(q_{df})(\omega^b - \omega_{df}^{df})$, and $\tilde{\omega}_{df} = \mathbf{R}_b^l(q_d)(\omega_{df}^{df} - \omega_d^d)$. The detail of the proof can be seen in [12].

With the definitions of (35) and (36), and according to (5), we re-express the rotation matrices as

$$\mathbf{R}_b^l(q) = \mathbf{R}_b^l(q_{df} \circ \tilde{q}) = \mathbf{R}_b^l(\tilde{q}) \mathbf{R}_b^l(q_{df}) \quad (39)$$

$$\mathbf{R}_b^l(q_{df}) = \mathbf{R}_b^l(q_d \circ \tilde{q}_{df}) = \mathbf{R}_b^l(\tilde{q}_{df}) \mathbf{R}_b^l(q_d) \quad (40)$$

Using (2) and (39) obtains

$$\begin{aligned} \delta_{3a} &= -\frac{T_{df}}{\hat{m}} [\mathbf{R}_b^l(q) - \mathbf{R}_b^l(q_{df})] \hat{e}_3 \\ &= -\frac{T_{df}}{\hat{m}} [\mathbf{R}_b^l(\tilde{q}) - \mathbf{I}] \mathbf{R}_b^l(q_{df}) \hat{e}_3 \\ &= -2 \frac{T_{df}}{\hat{m}} [\tilde{\eta} \mathbf{I} + S(\tilde{\varepsilon})] S(\tilde{\varepsilon}) \mathbf{R}_b^l(q_{df}) \hat{e}_3 \\ &= 2 \frac{T_{df}}{\hat{m}} (\tilde{\eta} \mathbf{I} + S(\tilde{\varepsilon})) S(\mathbf{R}_b^l(q_{df}) \hat{e}_3) \tilde{\varepsilon} \end{aligned}$$

Hence,

$$\delta_{3a} = \mathbf{W} \tilde{\varepsilon} \quad (41)$$

where $\mathbf{W} = 2 \frac{T_{df}}{\hat{m}} (\tilde{\eta} \mathbf{I} + S(\tilde{\varepsilon})) S(\mathbf{R}_b^l(q_{df}) \hat{e}_3)$. In addition, δ_{3b} is also derived in the same way as

$$\delta_{3b} = \mathbf{W}_{df} \tilde{\varepsilon}_{df} \quad (42)$$

where $\mathbf{W}_{df} = 2 \frac{T_d}{\hat{m}} (\tilde{\eta}_{df} \mathbf{I} + S(\tilde{\varepsilon}_{df})) S(\mathbf{R}_b^l(q_d) \hat{e}_3)$.

Now, continue the backstepping process and let the Lyapunov function be $V_3 = V_2 + 2(1 - \tilde{\eta}) + 2(1 - \tilde{\eta}_{df}) + 0.5 \delta_{3c}^2$. Differentiating V_3 with respect to time yields

$$\begin{aligned}\dot{V}_3 = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 + \delta_2^T[\delta_{3a} + \delta_{3b} + \delta_{3c}R_b^I(q_{df})\hat{e}_3 + \tilde{\Lambda}_a] \\ & + \tilde{e}^T R_b^I(q_{df})(\omega^b - \omega_{df}^{df}) + \tilde{e}_{df}^T R_b^I(q_d)(\omega_{df}^{df} - \omega_d^d) \\ & + \delta_{3c}(-\dot{T}_{df}/\hat{m} + \dot{T}_d/\hat{m}) \\ = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 + \tilde{e}^T[W^T\delta_2 + R_b^I(q_{df})(\omega^b - \omega_{df}^{df})] \\ & + \tilde{e}_{df}^T[W_{df}^T\delta_2 + R_b^I(q_d)(\omega_{df}^{df} - \omega_d^d)] \\ & + \delta_{3c}[\delta_2^T R_b^I(q_{df})\hat{e}_3 - \dot{T}_{df}/\hat{m} + \dot{T}_d/\hat{m}] + \delta_2^T \tilde{\Lambda}_a\end{aligned}$$

Next, we propose the following virtual control law

$$\omega^b = \Phi_3 = \omega_{df}^{df} - R_b^I(q_{df})k_{3a}\tilde{e} \quad (43)$$

$$\omega_{df}^{df} = R_b^I(q_d)^{-1}(-k_{3b}\tilde{e}_{df}) \quad (44)$$

$$\dot{T}_{df} = \hat{m}[\delta_2^T R_b^I(q_{df})\hat{e}_3 + k_{3c}\delta_{3c}] \quad (45)$$

which leads to the time derivative of V_3 as

$$\begin{aligned}\dot{V}_3 = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 - k_{3a}\tilde{e}^T\tilde{e} + \tilde{e}^T W^T\delta_2 - k_{3b}\tilde{e}_{df}^T\tilde{e}_{df} \\ & + \tilde{e}_{df}^T W_{df}^T\delta_2 - \tilde{e}_{df}^T R_b^I(q_d)\omega_d^d - k_{3c}\delta_{3c}^2 + \delta_{3c}\frac{\dot{T}_d}{\hat{m}} + \delta_2^T \tilde{\Lambda}_a\end{aligned}$$

Step 4: Introduce the Lyapunov command filter in the backstepping in Step 3 and define the backstepping error of the body rotational velocity as

$$\delta_4 = \omega^b - \Phi_3 = \omega^b - \Phi_{3f} + \Phi_{3f} - \Phi_3 = \delta_{4a} + \delta_{4b} \quad (46)$$

where $\delta_{4a} = \omega^b - \Phi_{3f}$ and $\delta_{4b} = \Phi_{3f} - \Phi_3$. Substituting (43) and (46) into (37), one obtains

$$\dot{\eta} = -\frac{1}{2}\tilde{e}^T[-k_{3a}\tilde{e} + R_b^I(q_{df})(\delta_{4a} + \delta_{4b})] \quad (47)$$

In this step, we choose the Lyapunov function candidate $V_4 = V_3 + 0.5\delta_{4a}^T\delta_{4a} + 0.5\delta_{4b}^T\delta_{4b}$ and derive the time derivative of V_4 as follows;

$$\begin{aligned}\dot{V}_4 = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 - k_{3a}\tilde{e}^T\tilde{e} + \tilde{e}^T W^T\delta_2 \\ & + \tilde{e}^T R_b^I(q_{df})(\delta_{4a} + \delta_{4b}) - k_{3b}\tilde{e}_{df}^T\tilde{e}_{df} + \tilde{e}_{df}^T W_{df}^T\delta_2 \\ & - \tilde{e}_{df}^T R_b^I(q_d)\omega_d^d - k_{3c}\delta_{3c}^2 + \delta_{3c}\frac{\dot{T}_d}{\hat{m}} + \delta_2^T \tilde{\Lambda}_a \\ & + \delta_{4a}^T[J^{-1}(-S(\omega^b)J\omega^b + \tau) + \Lambda_\alpha - \dot{\Phi}_{3f}] \\ & + \delta_{4f}^T[\dot{\Phi}_{3f} - \dot{\Phi}_3] \\ = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 - k_{3a}\tilde{e}^T\tilde{e} + \tilde{e}^T W^T\delta_2 \\ & - k_{3b}\tilde{e}_{df}^T\tilde{e}_{df} + \tilde{e}_{df}^T W_{df}^T\delta_2 - \tilde{e}_{df}^T R_b^I(q_d)\omega_d^d - k_{3c}\delta_{3c}^2 \\ & + \delta_{3c}\frac{\dot{T}_d}{\hat{m}} + \delta_2^T \tilde{\Lambda}_a + \delta_{4a}^T[R_b^I(q_{df})\tilde{e} + \Lambda_\alpha - \dot{\Phi}_{3f}] \\ & + J^{-1}(-S(\omega^b)J\omega^b + \tau) + \delta_{4b}^T[R_b^I(q_{df})\tilde{e} + \dot{\Phi}_{3f} - \dot{\Phi}_3]\end{aligned}$$

Finally, we select the following control law

$$\tau = \Phi_4 = S(\omega)J\omega + J[-R_b^I(q_{df})\tilde{e} - \hat{\Lambda}_\alpha + \dot{\Phi}_{3f} - k_{4a}\delta_{4a}] \quad (48)$$

$$\dot{\Phi}_{3f} = -R_b^I(q_{df})\tilde{e} - k_{4b}\delta_{4b} \quad (49)$$

such that the time derivative of V_4 becomes

$$\begin{aligned}\dot{V}_4 = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 - k_{3a}\tilde{e}^T\tilde{e} + \tilde{e}^T W^T\delta_2 - k_{3b}\tilde{e}_{df}^T\tilde{e}_{df} \\ & + \tilde{e}_{df}^T W_{df}^T\delta_2 - \tilde{e}_{df}^T R_b^I(q_d)\omega_d^d - k_{3c}\delta_{3c}^2 + \delta_{3c}\frac{\dot{T}_d}{\hat{m}} + \delta_2^T \tilde{\Lambda}_a \\ & - k_{4a}\delta_{4a}^T\delta_{4a} + \delta_{4a}^T \tilde{\Lambda}_\alpha - k_{4b}\delta_{4b}^T\delta_{4b} - \delta_{4b}^T \dot{\Phi}_3\end{aligned} \quad (50)$$

3.4 Closed-Loop Stability Analysis and Parameter Updating

In order to analyze the closed-loop stability of the overall system with control law (48), we select the overall Lyapunov function as

$$\begin{aligned}V = & V_4 + \frac{1}{2}\gamma_1\tilde{v}^T\tilde{v} + \frac{1}{2\lambda_1}tr[\tilde{W}_a^T\tilde{W}_a] + \frac{1}{2\lambda_2}\tilde{\alpha}_a^T\tilde{\alpha}_a + \frac{1}{2\lambda_3}\tilde{\beta}_a^T\tilde{\beta}_a \\ & + \frac{1}{2\lambda_4}\tilde{c}_a^T\tilde{c}_a + \frac{1}{2}\gamma_2\tilde{\omega}^{bT}\tilde{\omega}^b + \frac{1}{2\lambda_5}tr[\tilde{W}_\alpha^T\tilde{W}_\alpha] + \frac{1}{2\lambda_6}\tilde{\alpha}_\alpha^T\tilde{\alpha}_\alpha \\ & + \frac{1}{2\lambda_7}\tilde{\beta}_\alpha^T\tilde{\beta}_\alpha + \frac{1}{2\lambda_8}\tilde{c}_\alpha^T\tilde{c}_\alpha\end{aligned} \quad (51)$$

where $\tilde{v} = v - \hat{v}$, $\tilde{\omega}^b = \omega^b - \hat{\omega}^b$, and $\hat{v}$, $\hat{\omega}^b$ are the estimations of the velocity in inertial axis and the body rotational velocity in body axis by using (14) and (16) with RFWNN approximating the uncertainties. Let

$$\hat{v} = k_5\tilde{v} + g\hat{e}_3 - \frac{T}{\hat{m}}R_b^I(q)\hat{e}_3 + \hat{\Lambda}_a \quad (52)$$

$$\hat{\omega}^b = k_6\tilde{\omega}^b + \hat{J}^{-1}[-S(\omega^b)J\hat{\omega}^b + \tau] + \hat{\Lambda}_\alpha \quad (53)$$

Therefore,

$$\dot{\tilde{v}} = -k_5\tilde{v} + \hat{\Lambda}_a \quad (54)$$

$$\dot{\tilde{\omega}}^b = -k_6\tilde{\omega}^b + \tilde{\Lambda}_\alpha \quad (55)$$

Subsequently, the time derivative of (51) becomes

$$\begin{aligned}\dot{V} = & -k_1\delta_1^T\delta_1 - k_2\delta_2^T\delta_2 - k_{3a}\tilde{e}^T\tilde{e} + \tilde{e}^T W^T\delta_2 - k_{4a}\delta_{4a}^T\delta_{4a} \\ & - k_{3b}\tilde{e}_{df}^T\tilde{e}_{df} + \tilde{e}_{df}^T W_{df}^T\delta_2 - \tilde{e}_{df}^T R_b^I(q_d)\omega_d^d - k_{3c}\delta_{3c}^2 \\ & + \delta_{3c}\frac{\dot{T}_d}{\hat{m}} - k_{4b}\delta_{4b}^T\delta_{4b} - \delta_{4b}^T \dot{\Phi}_3 + (\delta_2 + \gamma_1\tilde{v})^T \tilde{\Lambda}_a \\ & - k_5\tilde{v}^T\tilde{v} + (\delta_{4a} + \gamma_2\tilde{\omega}^b)^T \tilde{\Lambda}_\alpha - k_6\tilde{\omega}^{bT}\tilde{\omega}^b \\ & + \frac{1}{\lambda_1}tr[\tilde{W}_a^T\dot{\tilde{W}}_a] + \frac{1}{\lambda_2}\tilde{\alpha}_a^T\dot{\tilde{\alpha}}_a + \frac{1}{\lambda_3}\tilde{\beta}_a^T\dot{\tilde{\beta}}_a + \frac{1}{\lambda_4}\tilde{c}_a^T\dot{\tilde{c}}_a \\ & + \frac{1}{\lambda_5}tr[\tilde{W}_\alpha^T\dot{\tilde{W}}_\alpha] + \frac{1}{\lambda_6}\tilde{\alpha}_\alpha^T\dot{\tilde{\alpha}}_\alpha + \frac{1}{\lambda_7}\tilde{\beta}_\alpha^T\dot{\tilde{\beta}}_\alpha + \frac{1}{\lambda_8}\tilde{c}_\alpha^T\dot{\tilde{c}}_\alpha\end{aligned} \quad (56)$$

From (22), it follows that

$$\tilde{\Delta}_a = \tilde{W}_a^T \hat{\phi}_a + \hat{W}_a^T \frac{\partial \varphi_a}{\partial \alpha_a} \tilde{\alpha}_a + \hat{W}_a^T \frac{\partial \varphi_a}{\partial \beta_a} \tilde{\beta}_a + \hat{W}_a^T \frac{\partial \varphi_a}{\partial c_a} \tilde{c}_a + \varepsilon_a \quad (57)$$

$$\tilde{\Delta}_\alpha = \tilde{W}_\alpha^T \hat{\phi}_\alpha + \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \alpha_\alpha} \tilde{\alpha}_\alpha + \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \beta_\alpha} \tilde{\beta}_\alpha + \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial c_\alpha} \tilde{c}_\alpha + \varepsilon_\alpha \quad (58)$$

The substitution of (57) and (58) into (56) leads to find the time derivative of (51) as

$$\begin{aligned} \dot{V} = & -k_1 \delta_1^T \delta_1 - k_2 \delta_2^T \delta_2 - k_{3a} \tilde{\varepsilon}^T \tilde{\varepsilon} + \tilde{\varepsilon}^T W^T \delta_2 - k_{4a} \delta_{4a}^T \delta_{4a} \\ & - k_{3b} \tilde{\varepsilon}_{df}^T \tilde{\varepsilon}_{df} + \tilde{\varepsilon}_{df}^T W_{df}^T \delta_2 - \tilde{\varepsilon}_{df}^T R_b^T(q_d) \omega_d^d - k_{3c} \delta_{3c}^2 \\ & + \delta_{3c} \frac{\dot{T}_d}{\hat{m}} - k_{4b} \delta_{4b}^T \delta_{4b} - \delta_{4b}^T \dot{\Phi}_3 \\ & + \frac{1}{\lambda_1} \text{tr} \left\{ \tilde{W}_a^T \left[\dot{W}_a + \lambda_1 \hat{\phi}_a U_1^T \right] \right\} \\ & + \frac{1}{\lambda_2} \left(\dot{\alpha}_a^T + \lambda_2 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial \alpha_a} \right) \tilde{\alpha}_a \\ & + \frac{1}{\lambda_3} \left(\dot{\beta}_a^T + \lambda_3 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial \beta_a} \right) \tilde{\beta}_a \\ & + \frac{1}{\lambda_4} \left(\dot{c}_a^T + \lambda_4 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial c_a} \right) \tilde{c}_a \\ & + \frac{1}{\lambda_5} \text{tr} \left\{ \tilde{W}_\alpha^T \left[\dot{W}_\alpha + \lambda_5 \hat{\phi}_\alpha U_2^T \right] \right\} \\ & + \frac{1}{\lambda_6} \left(\dot{\alpha}_\alpha^T + \lambda_6 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \alpha_\alpha} \right) \tilde{\alpha}_\alpha \\ & + \frac{1}{\lambda_7} \left(\dot{\beta}_\alpha^T + \lambda_7 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \beta_\alpha} \right) \tilde{\beta}_\alpha \\ & + \frac{1}{\lambda_8} \left(\dot{c}_\alpha^T + \lambda_8 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial c_\alpha} \right) \tilde{c}_\alpha \\ & - k_5 \tilde{v}^T \tilde{v} - k_6 \tilde{\omega}^b \tilde{\omega} + (\delta_2 + \gamma_1 \tilde{v})^T \varepsilon_a + (\delta_{4a} + \gamma_2 \tilde{\omega}^b)^T \varepsilon_\alpha \end{aligned} \quad (59)$$

in which $U_1 = \delta_2 + \gamma_1 \tilde{v}$, $U_2 = \delta_{4a} + \gamma_2 \tilde{\omega}^b$ where γ_1 and γ_2 are two positive real constants. The last eight terms in (59) can be eliminated if the following parameters updating rules are chosen

$$\begin{aligned} \dot{\hat{W}}_a &= -\dot{\tilde{W}}_a = \lambda_1 \hat{\phi}_a U_1^T, \quad \dot{\hat{\alpha}}_a^T = -\dot{\tilde{\alpha}}_a^T = \lambda_2 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial \alpha_a} \\ \dot{\hat{\beta}}_a^T &= -\dot{\tilde{\beta}}_a^T = \lambda_3 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial \beta_a}, \quad \dot{\hat{c}}_a^T = -\dot{\tilde{c}}_a^T = \lambda_4 U_1^T \hat{W}_a^T \frac{\partial \varphi_a}{\partial c_a} \\ \dot{\hat{W}}_\alpha &= -\dot{\tilde{W}}_\alpha = \lambda_5 \hat{\phi}_\alpha U_2^T, \quad \dot{\hat{\alpha}}_\alpha^T = -\dot{\tilde{\alpha}}_\alpha^T = \lambda_6 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \alpha_\alpha} \\ \dot{\hat{\beta}}_\alpha^T &= -\dot{\tilde{\beta}}_\alpha^T = \lambda_7 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial \beta_\alpha}, \quad \dot{\hat{c}}_\alpha^T = -\dot{\tilde{c}}_\alpha^T = \lambda_8 U_2^T \hat{W}_\alpha^T \frac{\partial \varphi_\alpha}{\partial c_\alpha} \end{aligned} \quad (60)$$

where $\lambda_i, i = 1 \sim 8$ denote the updating rates of the parameters of the RFWNN. With the parameter adjustment

laws, the time derivative of the Lyapunov function becomes

$$\begin{aligned} \dot{V} = & -k_1 \delta_1^T \delta_1 - k_2 \delta_2^T \delta_2 - k_{3a} \tilde{\varepsilon}^T \tilde{\varepsilon} + \tilde{\varepsilon}^T W^T \delta_2 - k_{4a} \delta_{4a}^T \delta_{4a} \\ & - k_{3b} \tilde{\varepsilon}_{df}^T \tilde{\varepsilon}_{df} + \tilde{\varepsilon}_{df}^T W_{df}^T \delta_2 - \tilde{\varepsilon}_{df}^T R_b^T(q_d) \omega_d^d - k_{3c} \delta_{3c}^2 \\ & + \delta_{3c} \frac{\dot{T}_d}{\hat{m}} - k_{4b} \delta_{4b}^T \delta_{4b} - \delta_{4b}^T \dot{\Phi}_3 - k_5 \tilde{v}^T \tilde{v} - k_6 \tilde{\omega}^b \tilde{\omega} \\ & + (\delta_2 + \gamma_1 \tilde{v})^T \varepsilon_a + (\delta_{4a} + \gamma_2 \tilde{\omega}^b)^T \varepsilon_\alpha \\ \leq & -k_1 \|\delta_1\|^2 - k_2 \|\delta_2\| \left(\|\delta_2\| - \frac{1}{k_2} \|\varepsilon_a\| \right) \\ & - k_{3a} \|\tilde{\varepsilon}\| \left(\|\tilde{\varepsilon}\| - \frac{1}{k_{3a}} \|W^T \delta_2\| \right) \\ & - k_{4a} \|\delta_{4a}\| \left(\|\delta_{4a}\| - \frac{1}{k_{4a}} \|\varepsilon_\alpha\| \right) \\ & - k_{3b} \|\tilde{\varepsilon}_{df}\| \left[\|\tilde{\varepsilon}_{df}\| - \frac{1}{k_{3b}} (\|W_{df}^T \delta_2\| + \|\omega_d^d\|) \right] \\ & - k_{3c} \delta_{3c} \left(\delta_{3c} - \frac{1}{k_{3c}} \frac{\dot{T}_d}{\hat{m}} \right) - k_{4b} \|\delta_{4b}\| \left(\|\delta_{4b}\| - \frac{1}{k_{4b}} \|\dot{\Phi}_3\| \right) \\ & - k_5 \|\tilde{v}\| \left(\|\tilde{v}\| - \frac{\gamma_1}{k_5} \|\varepsilon_a\| \right) - k_6 \|\tilde{\omega}^b\| \left(\|\tilde{\omega}^b\| - \frac{\gamma_2}{k_6} \|\varepsilon_\alpha\| \right) \end{aligned} \quad (61)$$

which is negative semi-definite if the following conditions are satisfied:

$$\begin{aligned} \|\delta_2\| &\geq \frac{1}{k_2} \|\varepsilon_a\|, \quad \|\tilde{\varepsilon}\| - \frac{1}{k_{3a}} \|W^T \delta_2\|, \quad \|\delta_{4a}\| \geq \frac{1}{k_{4a}} \|\varepsilon_\alpha\|, \\ \|\tilde{\varepsilon}_{df}\| &\geq \frac{1}{k_{3b}} (\|W_{df}^T \delta_2\| + \|\omega_d^d\|), \quad \delta_{3c} \geq \frac{1}{k_{3c} \hat{m}} \dot{T}_d, \quad \|\delta_{4b}\| \\ &\geq \frac{1}{k_{4b}} \|\dot{\Phi}_3\|, \quad \|\tilde{v}\| \geq \frac{\gamma_1}{k_5} \|\varepsilon_a\|, \quad \|\tilde{\omega}^b\| \geq \frac{\gamma_2}{k_6} \|\varepsilon_\alpha\| \end{aligned} \quad (62)$$

Theorem 1 If the parameters $k_1, k_2, k_{3a}, k_{3b}, k_{3c}, k_{4a}, k_{4b}, k_5, k_6, \gamma_1, \gamma_2, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6, \lambda_7, \lambda_8$ are positively real and the overall Lyapunov function is selected as in (51). By using the control laws, (24), (28), (43), and (48), the command filter described by (44), (45), and (49), the RFWNN estimator given by (18) and (19), and the parameter updating rules (60), then the overall closed-loop system with the system dynamics described in (13)–(16) is semi-globally uniformly ultimately bounded (SUUB). Moreover, all error variables $\delta_i, i = 1, \dots, 4$ converge to some neighborhood of the origin as $t \rightarrow \infty$.

Remark 1 All parameters of the RFWNN are updated online and do not need off-line training before implementation. The online supervised training is achieved by the comparison between the velocity and body rotation rate from the estimations of (52) and (53) and from the sensor. Hence, the parameter updating law (60) is a more generalized gradient descend rule by having the same formula

but different physical meanings on the error terms, U_1 and U_2 , by comparing to the classical gradient descend method.

Remark 2 The parameters, k_1 , k_2 , k_{3a} , and k_{4a} , are classical backstepping control gains, which determine the convergence rate of the states to the reference commands and the robustness of the classical backstepping controller. On the other hand, k_{3b} , k_{3c} , k_{4b} are the estimator and command filter gains dominate the convergence rate of the reference commands to the actual but unknown backstepping commands. The higher the filter gains, the more aggressive trajectories can be tracked. However, these result in large control commands of the actuators in quadrotors. Hence, k_5 and k_6 are designed to strengthen estimation stability; γ_1 and γ_2 increase one-degree-of-freedom parameter so as to adjust the ratios of the tracking errors and RFWNN approximation errors, in order to achieve the optimal turning parameters.

Table 1 Parameters of the used quadrotor

Parameter	Value	Parameter	Value
m (kg)	0.952	l_{beam} (m)	0.22
J_x (kg-m ²)	0.013215	I_{motor} (kg-m ²)	3.3×10^{-5}
J_y (kg-m ²)	0.012522	k_m (N-sec ²)	1.3×10^{-5}
J_z (kg-m ²)	0.023527	b (N-m-sec ²)	4.3×10^{-7}

Table 2 Parameters of the control law

Parameter	Value	Parameter	Value
k_1	1.5	k_{4b}	20
k_2	1	k_5	10
k_{3a}	1	k_6	10
k_{3b}	20	γ_1	5
k_{3c}	20	γ_2	5
k_{4a}	1	$\lambda_i, i = 1 \sim 8$	4

Table 3 Model uncertainties

Parameter	Value	Parameter	Value
Δm	10% of m	Δr_1	[0.003, 0, -0.02] (meter)
ΔJ_x	10% of J_x	Δr_2	[-0.001, 0, -0.02] (meter)
ΔJ_y	8% of J_y	Δr_3	[0, 0, -0.019] (meter)
ΔJ_z	6% of J_z	Δr_4	[0, -0.002, -0.02] (meter)
$\Delta \mathbf{x}_{cg}$	[0, 0, 0.02] (meter)	$\mathbf{F}_a$	[0.1, -1, 0] (N)
m	0.952 kg $\rightarrow$ 0.652 kg (at 20 s)		

4 Simulations and Discussion

In this section, four comparative simulations are conducted to examine the effectiveness and merit of the proposed controller. MATLAB/Simulink codes are used to do these four simulations where the fourth-order Runge–Kutta numerical method is employed with a fixed step size of 0.001 s. In the four simulations, Table 1 lists the used quadrotor parameters, and the quadrotor's position is initially set at the origin. There are two RFWNNs in the

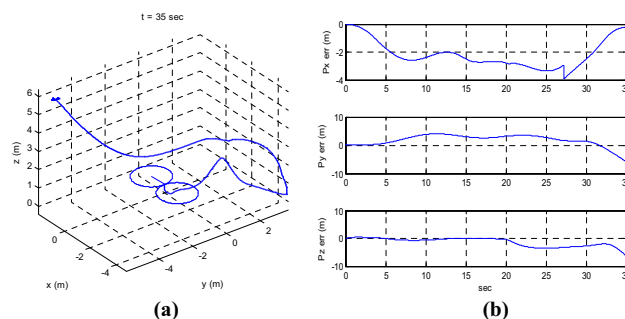


Fig. 3 Simulated results of the quadrotor tracking the eight-like trajectory and doing regulation by using the controller without RFWNN; **a** simulated trajectories of the quadrotor (bolder line) and the desired trajectory (thinner line) in meters; **b** tracking errors of the quadrotor in the x -, y -, and z -axes in meters

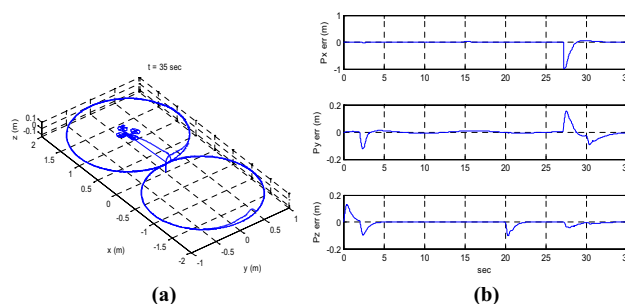
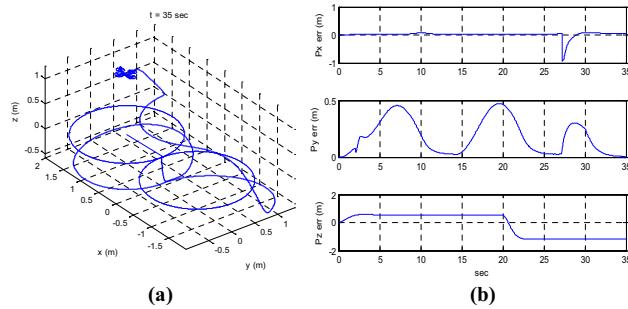


Fig. 4 Simulated results of the quadrotor tracking the eight-like trajectory doing regulation by using the controller using RFWNN. **a** simulated trajectories of the quadrotor (bolder line) and the desired trajectory (thinner line) in meters; **b** tracking errors of the quadrotor in the x -, y -, and z -axes in meters

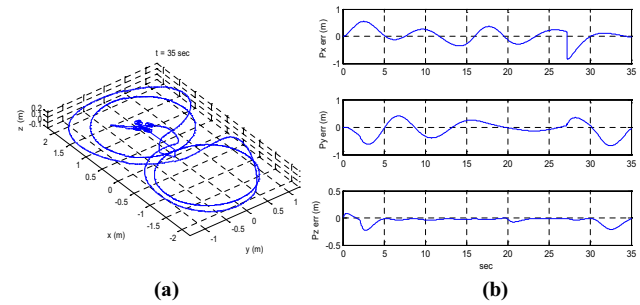
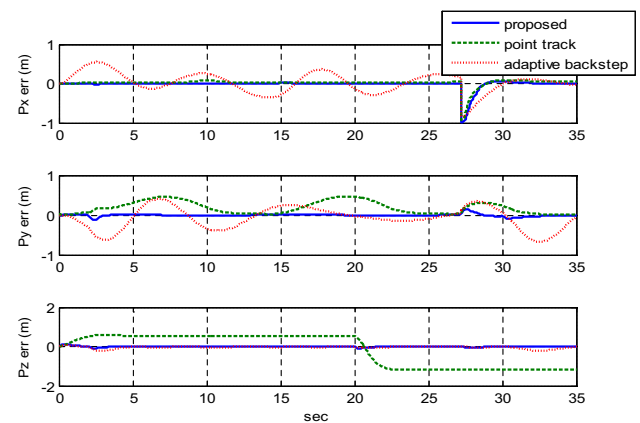
Table 4 Performance indices for tracking errors

Controller	Performance index			
	MSE	ISE	IAE	ITAE
Proposed controller without RFWNN	16.8805	590.8359	206.6312	4241.948
Proposed controller with RFWNN	0.017	0.5958	1.9872	43.6748
Improvement	99.9%	99.9%	99.04%	98.97%

**Fig. 5** Simulated results of the quadrotor tracking the eight-like trajectory doing regulation by using the controller proposed by Choi and Ahn [8]: **a** simulated trajectories of the quadrotor (bolder line) and the desired trajectory (thinner line) in meters; **b** tracking errors of the quadrotor in the x -, y -, and z -axes in meters

controller: The first one is used to online approximate the force uncertainties and has 12 nodes in the input layer, 5 neurons in the hidden layer, 3 nodes in the output layer, and the second one is employed to online learn the torque uncertainties and has 12 inputs, 8 neurons in the hidden layer, and 3 outputs. All parameters of RFWNNs are initialized with random numbers within the range of $[0, 0.01]$. Table 2 lists the parameters of the proposed control law, and Table 3 depicts the model uncertainties used for the simulations.

The first simulation is performed to test the proposed quaternion-based backstepping controller without RFWNN learning the uncertainties adaptively. The trajectory command takes an initial rest at the origin and starts to move along an eight-like trajectory that has two circles with diameters of 2 meters at the 2nd second. The first circle gets started at the origin with positive x -axis coordinates in the counterclockwise direction, whereas the second circle starts up at the origin in the clockwise direction, after the first circle being finished at the origin. Thus, the desired trajectories look like a drawing of writing an eight on the horizontal plane, thereby obtaining so-called eight-like trajectory. After the two circular tracking, the command stops at the origin and finally achieves regulation to the point, $[1, 0, 0]$. Moreover, the tuning rate of the trajectory is 0.5 rad/s and a steplike wind disturbance is added at the 30th second with the strength of 1 N in the direction of

**Fig. 6** Simulated results of the quadrotor tracking the eight-like trajectory doing regulation by using the controller proposed by Dong and Farrell [13]: **a** simulated trajectories of the quadrotor (bolder line) and the desired trajectory (thinner line) in meters; **b** tracking errors of the quadrotor in the x -, y -, and z -axes in meters**Fig. 7** Three directional tracking errors (in meter) of the quadrotor from the results of the proposed method (solid line in the 2nd simulation), nonlinear point tracking [8] (dashed line in the 3rd simulation), and command filtered adaptive backstepping [13] (dotted line in the 4th simulation)

$[0.0995, -0.995, 0]$. There is also a 0.3 kg (31.5%) mass changing which occurs at 20th second. After completing the eight-like trajectory at 27.2 s, a step command is given from origin to $[1, 0, 0]$ and the quadrotor will stay at this point until the end of the simulation. Figure 3a depicts the desired trajectory and simulated flight path of the quadrotor, and Fig. 3b displays the tracking errors of the overall closed-loop system. The simulation results show that the

Table 5 Performance index on tracking errors of 2nd, 3rd and 4th simulations

Controller	Performance index			
	MSE	ISE	IAE	ITAE
Proposed method	0.017	0.5958	1.9872	43.6748
Nonlinear point tracking [8]	0.7616	26.6563	34.8288	705.5851
Command filtered adaptive backstepping [13]	0.1545	5.4067	16.7298	285.4672
Improvement compared to [8]	97.77%	97.77%	94.29%	93.81%
Improvement compared to [13]	88.98%	88.98%	88.12%	84.7%

quadrotor is unable to track the trajectory due to small control gains and comparatively big uncertainties.

The second simulation, by comparing to the first simulation, is aimed to verify the performance of the proposed quaternion-based adaptive backstepping RFWNN controller for the uncertain quadrotor whose uncertainties are given in Table 3. The trajectory command is the same to those in the first simulation. The simulation results are shown in Fig. 4; the results in Fig. 4a illustrate the desired trajectory and simulated flight path of the quadrotor, and the tracking errors are depicted in Fig. 4b. Furthermore, the performance indices of the first two simulations are listed in Table 4. The results in Table 4 reveal that the tracking and regulation performance of the proposed controller using RFWNN is significantly improved.

In the following two comparative simulations, the proposed control method will be shown superior than the methods in [8, 13] do. This is because the proposed method can overcome the bad effects caused by parameter variations and disturbance. The same simulation settings are used in the third and the fourth simulations, but, at the third simulation, the control method proposed by Choi and Ahn [8] is adopted. Figure 5 depicts the simulation results; Fig. 5a shows the desired trajectory command and the actual trajectory of the controlled quadrotor; Fig. 5b displays the trajectory errors. It can be obviously seen from the results that the presented nonlinear point tracking control law by Choi and Ahn [8] is still unable to compensate for the effects of the uncertainties especially for the mass variations.

The fourth simulation is carried out to compare the proposed control method and a command filtered adaptive backstepping proposed by Dong et al. [13], in which the authors used a Lyapunov-based first-order command filter and an adaptive model with linearized parameters. The simulation results are presented in Fig. 6 where Fig. 6a shows the desired and actual moving trajectories of the quadrotor, and Fig. 6b plots the tracking errors. The results indicate that the adaptive control law has better performance in rejecting uncertainties and disturbances than the controller proposed by Choi and Ahn [8] does. However, the command filtered adaptive backstepping method

encounters significant tracking errors by comparing to the proposed method in the second simulation. Figure 7 depicts the tracking errors of the 2nd, 3rd, and 4th simulations, and Table 5 lists the performance indexes of these three simulations and the performance improvement percentages made by our proposed control law. The results indicate that our proposed method significantly outperforms the two existing methods by Choi and Ahn [8] and Dong et al. [13].

5 Conclusions and Future Work

This paper has presented a quaternion-based adaptive backstepping control method using recurrent fuzzy wavelet neural network (RFWNN) for regulation and trajectory tracking of quadrotors subject to model uncertainties and disturbances. The un-modeling part, including mass variations, wind disturbances, and center-of-gravity change, has been online learned by using the RFWNN. The quaternion-based adaptive backstepping RFWNN controller has been proposed by combining backstepping, quaternion control, and the two RFWNN online learners. The Lyapunov stability theory has been used to show the semi-globally uniformly ultimately bounded (SUUB) closed-loop stability. Through the four computer simulations, the proposed control method has been shown effective in trajectory tracking control and regulation of quadrotors with model uncertainties and disturbances. Furthermore, the proposed control method has also been shown superior by comparing to the two existing methods by Choi and Ahn [8] and Dong et al. [13].

Some interesting topics for future work would work toward the following three directions. First, a formation control together with trajectory tracking and collision-obstacle avoidance would be derived for a group of quadrotors by extending the proposed control law. Second, several experiments would be conducted to further verify the proposed control law. Third, the proposed method would be extended to trajectory tracking control of autonomous helicopters although the dynamics of helicopters is more complicated than that of the quadrotors.

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References

- Kiyashko, M.A., Sholmov, A.K., Efremov, A.A.: Quadrotor mathematical and information model analysis. In: Proceedings of IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering, pp. 901–904 (2017)
- Feng, Y., Rabbath, C.A., Rakheja, S., Su, C.Y.: Adaptive controller design for generic quadrotor aircraft platform subject to slung load. In: Proceedings of 2015 IEEE 28th Canadian Conference on Electrical and Computer Engineering, pp. 1135–1139 (2015)
- Lucia, S.D., Tipaldi, G.D., Burgard, W.: Attitude stabilization control of an aerial manipulator using a quaternion-based backstepping approach. In: Proceedings of 2015 IEEE European Conference on Mobile Robots, pp. 1–6 (2015)
- Ariyanto, M., Paryanto, M., Naniwa, T.: Mathematical and physical modelling with dynamic change in the center of gravity of quadrotor. In: Proceedings of 2016 8th International Conference on Information Technology and Electrical Engineering (ICITEE), pp. 1–6 (2016)
- Lee, S., Giri, D.K., Son, H.: Modeling and control of quadrotor UAV subject to variations in center of gravity and mass. In: Proceedings of 2017 14th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), pp. 85–90 (2017)
- Tran, N.K., Bulka, E., Nahon, M.: Quadrotor control in a wind field. In: 2015 International Conference on Unmanned Aircraft Systems (ICUAS), pp. 320–328 (2015)
- Bialy, B.J., Klotz, J., Brink, K., Dixon, W.E.: Lyapunov-based robust adaptive control of a quadrotor UAV in the presence of modeling uncertainties. In: Proceedings of 2013 American Control Conference, pp. 13–18 (2013)
- Choi, Y.-C., Ahn, H.-S.: Nonlinear control of quadrotor for point tracking: actual implementation and experimental tests. IEEE/ASME Trans. on Mechatronics **20**(3), 1179–1192 (2015)
- Rashad, R., Aboudonia, A., El-Badawy, A.: Backstepping trajectory tracking control of a quadrotor with disturbance rejection. In: Proceedings of 2015 XXV International Conference on Information, Communication and Automation Technologies (ICAT), pp. 1–7 (2015)
- Lee, C.T., Tsai, C.C.: Nonlinear adaptive aggressive control using recurrent neural networks for a small scale helicopter. Mechatronics **20**(4), 474–484 (2010)
- Tsai, C.C., Lee, C.T., Hwang, K.S.: Intelligent Adaptive Trajectory Tracking Control Using Fuzzy Basis Function Networks for an Autonomous Small-Scale Helicopter. In: Proceedings of 2011 IEEE International Conference on Systems, Man, and Cybernetics (2011)
- Zhao, S., Dong, W., Farrell, J.A.: Quaternion-based trajectory tracking control of VTOL-UAVs using command filtered backstepping. In: Proceedings of 2013 American Control Conference, pp. 1018–1023 (2013)
- Dong, W., Farrell, J.A., Polycarpou, M.M., Sharma, M.: Command filtered adaptive backstepping. IEEE Trans. Control Systems Technology **20**, 566–580 (2012)
- Castillo, A., Sanz, R., Garcia, P., Albertos, P.: A quaternion-based and active disturbance rejection attitude control for quadrotor. In: Proceedings of 2016 IEEE International Conference on Information and Automation (ICIA), pp. 240–245 (2016)
- Zou, Y., Huo, W.: Adaptive backstepping trajectory tracking controller for a small-scaled helicopter. In: Proceedings of 2015 34th Chinese Control Conference (CCC), pp. 737–742 (2015)
- Zak, S.H.: System and control. Oxford University Press, Oxford (2003)
- Tong, S., Li, Y., Sui, S.: Adaptive fuzzy tracking control design for SISO uncertain non strict feedback nonlinear systems. IEEE Trans. Fuzzy Syst. **24**(6), 1441–1454 (2016)
- Li, Y., Liu, L., Feng, G.: Robust adaptive output feedback control approach to a class of non-triangular stochastic nonlinear systems. Automatica **89**, 325–332 (2018)
- Kuo, C.W., Tsai, C.C.: Integral Consensus Tracking Using RFWNN for Uncertain Networked Heterogeneous Second-Order Nonlinear Multiagent Systems. In: Proceedings of 2016 International Conference on Advanced Robotics and Intelligent Systems, Taipei, Taiwan (2016)



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